

Technologies Cheat Sheet for Recruiters By Laxminarayana

Area of Expertise: Front End and UX

Technical Role	Skills Needed
.NET Developer	ASP, ASP .NET, C#, VB.NET
Cocoa Developer/OSX Developer	Apple, Apple script, Cocoa, Cocoa Touch, iOS, Objective-C, Objective C++, OS X, Xcode
Flash Developer	ActionScript, Flash
Front End Developer	ActionScript, Ajax, ASP, CoffeeScript, CSS, Flash, HTML, HTML5, JavaScript, JQuery
Interactive Developer	Ajax, CSS, HTML, JavaScript, PHP, XHTML
Rails Developer	Ruby, Ruby-HTML, Ruby on Rails
SharePoint Developer	Ajax, ASP .NET, C# , CSS, HTML, JavaScript, SharePoint, SQL, Windows Server, VB.NET, XML, XSL
Windows Developer	ASP, C, C#, C++, .NET, Windows

Area of Expertise: Back End

Technical Role	Skills Needed
Back End Developer	C#, Java, PHP, Python, Ruby
Business Intelligence Analyst	Business-intelligence, BI, C, C++, C#, Cognos, ETL, Java, jQuery, Objective-C, R, SQL, SAS, SSAS, SSIS, SSRS, Visual Basic, SAPUI5 - CSS, CSS3, HTML5, Javascript, OData Dynpro - ABAP, Dynpro
Data Engineer	C, C++, Cassandra, HBase, HDFS, Java, Puppet, Ruby, Scala
Data Mining	Perl, Python
Linux/Unix Proficient Developers	AWK, Debian, Emacs Fedora, Linux, Linux-kernel, RHEL, Unix, vi, vim Shell Scripting: Bourne Shell, Bash, Korn, C Shell Scripting: PHP, Python, Perl
Machine Learning/ Recommender Engines	Java, Perl, Python, R, Weka
Oracle Developer	ASP, .NET, SQL Databases: MySQL, Oracle, PostgreSQL, RDBMS
SAP Developer	ABAP, C++, COBOL, ERP, mySAP, Netweaver, Java, SAP
Site Reliability Engineer	Authentication/Authorization, Cloud, HTTP, Linux/Unix, SQL, Network Design, Security, performance, and disaster recovery, SSL, TCP/IP (languages will vary by company)

Software Architect	.NET, Agile, Ant, API, C++, Cassandra, CORS, CouchDB, Git, HTTP, Java, Maven, Mongo, MySQL, MVC, Oracle, Perl, PHP, Python, Redis, Ruby, SOA, SDK, Teradata, Unit-testing, Websocket (Vary by company, search on languages, frameworks, APIs, tools, databases, etc.)
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Area of Expertise: Big Data

Technical Role	Skills Needed
Big Data Engineers	Cassandra, Cloud Computing, Hadoop, HBase, MapReduce, Mongo
Data Engineer	C, C++, Cassandra, HBase, HDFS, Java, Puppet, Ruby, Scala
Data Mining	Perl, Python
Machine Learning/ Recommender Engines	Java, Perl, Python, R, Weka
Oracle Developer	ASP, .NET, SQL Databases: MySQL, Oracle, PostgreSQL, RDBMS
Platforms Infrastructure	C, C++, Cassandra, HDFS, Go, HTTP, Java, MySQL

Area of Expertise: Mobile

Technical Role	Skills Needed
All Mobile Development	Multithreading, Event-handling, Memory-management
Android Developer	C, C++, HTML5, Java, Linux
Blackberry Developer	C, C++, Cascades, HTML5, Java, QT
iOS Developer	C, Cocoa, HTML5, LUA, Objective-C
OS X Developer	Cocoa, Cocoa Touch, Objective-C, Objective-C++, OS X,

	Xcode
Web Developer	Ajax, ASP, CoffeeScript, CSS, HTML, JavaScript, JQuery, JRuby, PHP, Ruby, Ruby on Rails
Ruby on Rails Developer	Ruby, Ruby-HTML, Ruby on Rails
Interactive Developer	Ajax, CSS, HTML, JavaScript, PHP, XHTML
.NET Developer	ASP, ASP .NET, C#, VB.NET

Area of Expertise: Quality Assurance

Technical Role	Skills Needed
Build Release Engineer/Software Release Engineer/Gatekeeper	Scripts: Maven, Perl, Python, Shell, TCL Repository Tools: ClearCase, Git, Perforce, SVN (Subversion) Languages: C++, OO languages Build tools: Aardvark, Ant, Gradle, Hudson, Jenkins, Make, Mave

Frontend Languages, Backend Languages Frameworks, and skills mentioned below in a tabular format



Frontend languages, frameworks, & skills	Description
AJAX (Asynchronous JavaScript and XML)	A group of web development techniques used to create asynchronous applications. Asynchronous applications provide users with continuously updated app data, which means improved app performance and responsiveness.
Angular	A TypeScript-based free and open-source framework for building web applications, created and maintained by Google.
CSS (Cascading Style Sheets)	A style language that describes how a document (written in a markup language such as HTML) should be displayed. CSS tells the browser which HTML tags to style and how they should be formatted in terms of color, font, layout, etc.
HTML (HyperText Markup Language)	HTML defines the content and structure of webpages. HTML can also be used to embed images, text, and other objects into a webpage.
JavaScript	A programming language used in almost all webpages. It is a high-level, object-oriented, just-in-time language, often used for interactive webpage elements like dropdown menus or animated images.
JSON (JavaScript Object Notation)	A lightweight, open standard format for storing and transmitting data. JSON is often used when data is sent from a server to a webpage.
React	A JavaScript-based, free, and open-source library for building user interfaces, created and maintained by Facebook. Currently, React is the most popular library/framework for frontend development.
TypeScript	A programming language that builds on JavaScript. Typically, TypeScript is preferred for large-scale web applications, and JavaScript is used to develop interactive webpages.
UI (User Interface)	The space of interaction between humans and machines. User interfaces are designed to be intuitive and visually appealing.
UX (User Experience)	How a user interacts with a website, app, or other product. The best UX optimizes usability and overall experience.
Vue	A JavaScript-based, open-source framework for building user interfaces.

Backend languages, frameworks, & skills	Description
API (Application Programming Interface)	A connection between software programs. The term API can refer to either (1) the specification of how software components should interact with each other, or (2) the implementation of the API's service.
REST (Representational State Transfer) API	A software architectural style designed to make the web more scalable. This is the predominant web API model.
SOAP (Simple Object Access Protocol) API	A messaging protocol specification for exchanging structured data in the implementation of web services.
ASP.net	A web framework (part of the .NET framework) used for building dynamic web applications. It is frequently used with Microsoft products such as C#.
C	One of the original compiled, general-purpose programming languages. It is still used widely in mostly low-level applications such as operating systems and hardware drivers.
C++	An object-oriented, compiled, general-purpose programming language designed as an improvement on C. Its popularity has declined somewhat. However, it is still used in applications that require high performance, such as servers and search systems.
C#	An object-oriented compiled programming language designed as an improvement on C++. Developed by Microsoft in 2000, it is used with Windows applications, in many large enterprises, and by gaming companies.
Django	An open-source Python web application framework.
Flask	An open-source Python web application framework.
Go (or Golang)	A statically typed, compiled programming language designed by Google.
Java	An object-oriented, compiled, general purpose programming language developed by Oracle. It is often used in enterprise software.
MySQL	An open source relational database management system, and the most widely used open source SQL database. It is currently owned by Oracle.
Node	An open-source, cross-platform, runtime environment designed to run JavaScript as a server-side language. Node has an event-driven architecture that enhances scalability to concurrent users.
NoSQL (not only SQL)	A database for storing and managing data on a large scale. Popular examples include MongoDB, CouchDB, CouchBase, Cassandra, and Redis
PHP	An open-source, server-side scripting language designed for web development. It was popular in the 1990s and early 2000s, but today it's considered outdated.
Python	An open-source interpreted, object-oriented, general-purpose programming language. Python is currently the most popular language for tech companies and scientific research.
Ruby	An open-source interpreted, object-oriented, general-purpose programming language designed for fast development time. Though it peaked in popularity around 2008, it is still used by many tech companies and startups. Ruby is paired with the Rails framework (Ruby on Rails).
Scala	A statically typed, general-purpose programming language used for both object-oriented and functional programming. Scala is interoperable with Java and intended to be compiled on a Java Virtual Machine.
SDK (Software Development Kit)	A collection of software development tools provided by the manufacturer of an operating system, hardware platform, or language that allows for the creation of applications.
Spring	The most popular open-source framework for Java.

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DevOps skills	Description
Ansible	An open-source IaaS (infrastructure as a service) automation tool for cloud provisioning, configuration management, and application deployment. It is similar to Chef , Puppet , and Salt .
AWS (Amazon Web Services)	An on-demand cloud computing platform operated by Amazon. Currently, it is the most popular cloud computing web services platform.
Azure	A cloud computing platform operated by Microsoft. It's currently the second most popular cloud-computing web services platform.
Chef	A commercial tool for configuring, managing and deploying cloud infrastructure, similar to Ansible , Puppet , and Salt .
CI/CD (Continuous Integration/Continuous Delivery)	A software development practice of rapidly building, testing, and releasing software to users by frequently and incrementally introducing changes. Automation and continuous monitoring ensure that code is released quickly and reliably.
Docker	A PaaS (platform as a service) product for building, running, and managing containers on the cloud.
GCP (Google Cloud Platform)	A cloud computing platform. Currently, it is the third most popular cloud computing web services platform.
Git	Software used for distributed revision control and tracking and managing source code changes. It emphasizes speed and data integrity, allowing developers to track revisions and history and coordinate their work on a coding project. GitHub , BitBucket , GitLab , and SourceBucket offer Git repositories as a service.
Jenkins	An open-source automation software that enables developers to build, test and deploy software, facilitating CI/CD .
Kubernetes	An open-source system for automating containerized applications' deployment, scaling, and management.
Puppet	An automation and configuration management tool similar to Ansible , Chef , and Salt .
Salt	Open-source software used for event-driven IT automation and configuration management. It is similar to Ansible , Chef , and Puppet .
Terraform	An open-source IaaS (infrastructure as a service) automation tool for cloud provisioning, configuration management and service orchestration.

Popular DevOps roles	Description
Build/Release Engineer	Works with developers to test and deploy software. They are also involved with version control, automation, and configuration management.
Cloud Architect	Oversees the overall cloud computing strategy, converting technical requirements into the design, development, and management of cloud architecture and infrastructure
Infrastructure Engineer	Responsible for overall infrastructure, including design, security, virtualization, containerization, automation, deployment, and reliability.
Security (DevSecOps) Engineer	Incorporates security controls to protect information systems from misuse, fraud, information theft, and damage.
Site Reliability Engineer	Ensures scalability, reliability, and availability of software systems.
System Administrator (SysAdmins)	Works on the maintenance, configuration, and operation of computer systems and services, focusing on performance, security, and troubleshooting.
Quality Assurance/ Test Engineer	Ensures that software releases are reliable and tested to be error-free before being released to users.

Front-end developers

A front-end developer is responsible for the client-facing side of the product, whether that be a website, a mobile app, or the user interface of non-public software. They have to be able to communicate with both technical and non-technical departments within their organization, particularly product and design. Their titles will typically be something like **Front End Software Engineer or Developer, UI/UX Engineer, or JavaScript Developer.**

Soft skills to look for in front-end developers include:

- The ability to translate technical info into non-technical language
- Sensitivity to the needs of the end-user
- Good attention to detail
- Committed to professional development
- The ability to work as part of a team

Back-end developers

A back-end developer creates the foundation for software and web applications that is invisible to the end user. They're responsible for the essential background operations of a product like storage and retrieval of data, application architecture and infrastructure, and the scalability of the product. Their titles will typically be something like **Python/C/C++/C#/Java Developer, Member of Technical Staff, or Machine Learning Engineer.**

So what soft skills do these developers need?

- Problem-solving
- Adaptability
- Good communication
- Accountability
- Good team player
- A fast learner, particularly when it comes to new tech

Full-stack developers

A full-stack developer performs both front-end and back-end functions and tends to specialize in a few specific technologies. They will usually have more general titles to reflect their broader skill set; if they aren't referred to as a Full Stack Developer, then they'll probably just be called a Software Engineer or Developer.

They need soft skills like:

- Creative problem-solving
- Initiative
- Both tech and non-tech communication
- Good team working and collaboration
- A great multi-tasker with an all-round perspective

DevOps developers

DevOps specialists combine both development and operations. They're responsible for ensuring that products and updates are released quickly and efficiently by taking care of things like cloud infrastructure, security, and continuous integration & deployment. Their title might just be **DevOps engineer**, or it might be something more specialized like **Cloud Engineer** or **DevSecOps Engineer**.

Soft skills they need include:

- Strong leadership
- The ability to collaborate
- Problem-solving
- Excellent communication
- A track record of strategic decision-making
- Driven and proactive

Web Technologies – Java

Java, EJB, RMI, Swing, CORBA, Servlets, JSP, JDBC, Core Java, Struts framework, Web sphere, Web logic, HTML, XML, Java script, J2EE, J2ME, DOT NET, framework, LDAP, Internet Security, JVM, Pjava, MIPS, Tibco, Board Design, Ematrix, XSL, XSLT

Web Technologies – Microsoft

VB.NET, ASP.NET, ADO.NET, VC++.NET, C#, COM, DCOM

Testing

Win runner, Load runner, Telecom Testing, White box testing, AIX testing ' Testcases, Test director, Test Suite, Silk Test, Mercury Testing, rational Robo,

Quality

Professionals with knowledge of ISO9000, SEICMM

Systems Software

C/C++ on Unix/Linux, DSP, Chip level designing, Unix kernel architecture, file system, memory management, RTOS, L2 support etc

Client Server

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VB, VC++, COM, DCOM with RDBMS – Oracle, SQL, Sybase

Embedded Systems

RTOS, C on UNIX, Networking, Vx works, design and code reviews, QNX, Net kernel, Psos, IRMK, Firmware, BIOS, Assembly, MPEG DVD, MP3, JPEG, Set Top Box, DVB, Microprocessors, Microcontrollers, ADA, AVIONICS, D0178B, Industrial Automation,

Mainframes

IBM Mainframe, AS/400, OS/390, z/OS, OS/400, VAX/VMS Mainframe, COBOL, CICS, MVS, DB2, IDMSX, IMS, Tandem, Natural ADABAS, Xpeditior, RPG, JCL

Networking

Networking S/W, UNIX System Admin., Windows NT Admin, TCP/IP, SPX, X25, Network Management – SNMP, CMIP, NMS, LAN/WAN, ATM, MPLS, FR, ISDN, Routing Protocols – BGP OSPF, RTP, RIP, IPV6, SONET, SDH

ERP / SAP

ERP and SAP Implementations of all modules, SAP-SD/MM/PP, FICO, ABAP4, Basis, People Soft-HRM, finance Module, BAN, Oracle Manufacturing, CRM-Siebel /Clarify

TELECOM

Switching, GSM, GPRS, UMTS, 3G, Layer-C, RLC, RRC, NodeB, DSP, TDMA, CDMA, WAP, SS7, ATM, FR, VOIP-SIGTRAN, MGCP, MEGACO, SIP, H.323, 245, TMN, Signalling, Bluetooth, GPS, Home Networking-JINI, HAVI.

C, UNIX

Device Drivers, X-Motif, Linux, Linux, /NT/VMS, Internalls, kernels, TCL/TK, Multithreading, Socket programming, Storage Device (Isctsi, SCSI, SAN, Veritas, Volume Manager, RAID, Fiber Channel, NAS)

VC++

DCOM, ATL, OOAD, XML, WDM, Vxd, WinSDK, WINCE, Palm OS, EPOC, WIN 32, API, Device Drivers, X-Windows, Direct X, Active X, NET, C#.

HARDWARE

ASIC, VLSI, FPGA, Verilog, VHDL, XiLINK, Physical Design, CMOS, Design, IC Design, Place & Route, Synthesis, EDA Tools, FPGA Design, Board Design, PCB Design, Circuit Design, Mixed Signal, Analog Design

APPLICATION

Oracle 8i, 11i, Oracle DBA, SQL-DBA, PB, Delphi, Data Modeling, Data warehousing – OLAP, ROLAP, Oracle DBA, VB, ASP, COM, DCOM, NET, C#, Oracle , PL/SQL, Pro*C, D2K

CAD/CAM

CAD, CAE, CAM, Catia V4,V5, Pro-E, Uni graphics, Hyper mesh, IDEAS, NASTRAN, ANSYS, Auto cad Tools, SQA, Quality, Testing / Implementation, Level 4 / Level 5 / ISO / Six Sigma / Technical writers, architects , Lotus Notes, System Administrator, System Administrator, UNIX / LINUX, SUN SOLARIS

Brief Information about Technologies and

Abbreviations Web Technologies

Java

An object-oriented programming language that is platform independent (thesame Java program runs on all hardware platforms without modification). Developed by Sun, Java is widely used on the Web for both client and server processing. Modeled after C++, Java added programming enhancements such as "garbage collection," which automatically frees unused memory. It was also designed to run in small amounts of memory. The first Web browsers to run Java were Sun's HotJava and Netscape Navigator 2.0.

EJB

(**Enterprise JavaBeans**) A software component in Sun's J2EE platform, which provides a pure Java environment for developing and running distributed applications. EJBs are written as software modules that contain the business logic of the application. They reside in and are executed in a runtime engine called an "EJB Container," which provides a host of common interfaces and services to the EJB, including security and transaction support. At the wire level, EJBs look like CORBA components.

RMI

(**Remote Method Invocation**) A standard from Sun for distributed objects written in Java. RMI is a remote procedure call (RPC), which allows Java objects (software components) stored in the network to be run remotely. Unlike CORBA and DCOM objects, which can be developed in different languages, RMI is designed for objects written only in Java.

SWING

A Java toolkit for developing graphical user interfaces (GUIs). It includes elements such as menus, toolbars and dialog boxes. Swing is written in Java and is thus platform independent, unlike the Java Abstract Window Toolkit (AWT), which provides platform-specific code. Swing also has more sophisticated interface capabilities than AWT and offers such features as

tabbed panes and the ability to change images on buttons. Swing is included in the Java Foundation Classes (JFC) which are provided in the Java Developers Toolkit (JDK).

COBRA

Common Object Request Broker Architecture) A software-based interface from the Object Management Group (OMG) that allows software modules (objects) to communicate with each other no matter where they are located on a private network or the global Internet. CORBA is a "distributed objects" system designed for multi-tier, client/server applications, where processing data in one computer requires additional processing by some other service in another computer in order to complete the transaction. CORBA is also described as an "object bus" or "software bus."

SERVELETS

A Java application that runs in a Web server or application server and provides server-side processing such as accessing a database and e-commerce transactions. Widely used for Web processing, servlets are designed to handle HTTP requests (get, post, etc.) and are the standard Java replacement for a variety of other methods, including CGI scripts, Active Server Pages (ASPs) and proprietary C/C++ plug-ins for specific Web servers (ISAPI, NSAPI).

JSP

(Java Server Page) An extension to the Java servlet technology from Sun that allows HTML to be combined with Java on the same page. The Java provides the processing, and the HTML provides the page layout that will be rendered in the Web browser

JDBC

(Java DataBase Connectivity) A programming interface that lets Java applications access a database via the SQL language. Since Java interpreters (Java Virtual Machines) are available for all major client platforms, this allows a platform-independent database application to be written. In 1996, JDBC was the first extension to the Java platform.

STRUTS

A framework for writing Web-based applications in Java that supports the Model-View-Controller (MVC) architecture. Struts is deployed as JSP pages using special tags from the Struts tag library, which includes routines for building forms, HTML rendering, storing and retrieving data and business

logic

WEB LOGIC

A software suite from BEA Systems, Inc., San Jose, CA (www.beasys.com) that is used to deploy Web and SOA applications. The core product is BEA WebLogic Server, a J2EE application server. BEA WebLogic Portal is an enterprise portal that offers advanced searching, and BEA WebLogic Integration provides tools for transforming and routing data from multiple sources. BEA WebLogic Enterprise is the integrated development environment (IDE) for the WebLogic family

HTML

HyperText Markup Language) The document format used on the Web. Web pages are built with HTML tags (codes) embedded in the text. HTML defines the page layout, fonts and graphic elements as well as the hypertext links to other documents on the Web. Each link contains the URL, or address, of a Web page residing on the same server or any server worldwide, hence "World Wide" Web.

XML

(EXtensible Markup Language) An open standard for describing data from the W3C. It is used for defining data elements on a Web page and business-to-business documents. XML uses a similar tag structure as HTML; however, whereas HTML defines how elements are displayed, XML defines what those elements contain.

While HTML uses predefined tags, XML allows tags to be defined by the developer of the page. Thus, virtually any data items, such as "product," "sales rep" and "amount due," can be identified, allowing Web pages to function like database records. By providing a common method for identifying data, XML supports business-to-business transactions and has become "the" format for electronic data interchange and Web services

J2EE

(Java 2 Platform, Enterprise Edition) A platform from Sun for building distributed enterprise applications. J2EE services are performed in the middle tier between the user's machine and the enterprise's databases and legacy information systems. J2EE comprises a specification, reference implementation and set of testing suites. Its core component is Enterprise JavaBeans (EJBs), followed by JavaServer Pages (JSPs) and Java servlets and a variety of interfaces for linking to the information resources in the enterprise.

J2ME

(Java 2 Platform, Micro Edition) A version of Java 2 for cellphones, PDAs and consumer appliances. J2ME uses the K Virtual Machine (KVM), a specialized Java interpreter for devices with limited memory. The Connected Limited Device Configuration (CLDC) provides the programming interface for wireless applications. The Mobile Information Device Profile (MIDP) provides support for a graphical interface, networking and storage.

DOT NET

(.NET) A comprehensive software development platform from Microsoft that was introduced in 2000 as the company's next generation programming environment. Pronounced "dot- **net**," and widely known as the ".NET Framework," it was designed to compete with the Java J2EE platform

LDAP

Lightweight Directory Access Protocol) A protocol used to access a directory listing. LDAP support is being implemented in Web browsers and e-mail programs, which can query an LDAP-compliant directory. It is expected that LDAP will provide a common method for searching e-mail addresses on the Internet, eventually leading to a global white pages. LDAP is a sibling protocol to HTTP and FTP and uses the ldap:// prefix in its URL.

JVM

A Java interpreter. The Java Virtual Machine (JVM) is software that converts the Java intermediate language (bytecode) into machine language and executes it. The original JVM came from the JavaSoft division of Sun. Subsequently, other vendors developed their own; for example, the Microsoft Virtual Machine is Microsoft's Java interpreter. A JVM is incorporated into a Web browser in order to execute Java applets. A JVM is also installed in a Web server to execute server-side Java programs. A JVM can also be installed in a client machine to run stand-alone Java applications

PJAVA

A version of Java from Sun intended for PDAs and other handheld devices. EmbeddedJava (EJava) is a counterpart set of technologies that provide support for character-based displays or devices without displays rather than graphical interfaces. PersonalJava (PJava) is intended for open systems that require Web browsing, and PJava includes applet support. EJava is intended for closed systems that have severe restrictions on memory.

MIPS

(Million Instructions Per Second) The execution speed of a computer. For example, .5MIPS is 500,000 instructions per second; 100 MIPS is a hundred million instructions per second. MIPS was a popular rating before computers reached gigahertz speeds, but MIPS rates were never uniform. Some were best-case mixes while others were averages. In addition, it takes more instructions in one machine to do the same thing as another (RISC vs. CISC, mainframe vs. PC). As a result, MIPS has been called "MisInformation to Promote Sales" as well as "Meaningless Interpretation of Processor Speed."

XSL

(eXtensible Stylesheet Language) A standard from the W3C for describing a style sheet for XML documents. It is the XML counterpart to the Cascading Style Sheets (CSS) in HTML and is compatible with CSS2. XSL is made up of three components:

(1) XSL Transformations (XSLT) is the processing language for XSL. It is used to convert XML documents into HTML or other document types and may be used independently of XSL. (2) XML Path Language (XPath) is used to identify and select tagged elements within an XML document, and (3) XSL Formatting Objects (XSL FO) provides the format vocabulary

XSLT

(eXtensible Stylesheet Language Transformation) Software that converts an XML document into another format such as HTML, PDF or text. It may also be used to convert one XML document to another XML document with a different set of XML tags (different schema).

VB.NET

(Visual Basic .NET) An object-oriented programming language from Microsoft. It is the .NET version of the Visual Basic (VB) programming language. Like all .NET languages, VB.NET uses the Common Language Runtime (CLR) for program execution. VB.NET is substantially different from traditional Visual Basic, which has been the most popular language for developing Windows applications

ASP

Application Service Provider) An organization that hosts software applications on its own servers within its own facilities. Customers rent the use of the application and access it over the Internet or via a private line connection. Also called a "commercial service provider." The Web browser, acting as a universal client interface, has fueled this "on-demand software" market.

ASP.NET

ASP.NET, also known as ASP+, is an enhanced version of ASP for the .NET platform. It supports executable programs compiled from C#, C++ and other languages and is not backward compatible with regular ASP code. ASP.NET pages are always compiled rather than interpreted as are ASP pages

ADO

(ActiveX Data Objects) A programming interface from Microsoft that is designed as "the" Microsoft standard for data access. First used with Internet Information Server, ADO is a set of COM objects that provides an interface to OLE DB. The three primary objects are Connection, Command and Recordset. The Connection object establishes a connection with a particular database management system (DBMS) or other data source. It can also send a query to the database. The Command object is an alternate way of sending a query to the database, and the Recordset object contains the resulting answer, which is a group of records

ADO.NET

ADO.NET is the .NET version of ADO, which is substantially different from ADO. It supports XML documents and relies on .NET Data Providers as an interface layer between the application and the databases.

TESTING WINRUNNER

Comprehensive automated application testing software for Windows from Mercury Interactive Corporation, Mountain View, CA (www.mercury.com/us). It lets users compare expected and actual outcomes and provides wizards for automatically setting up tests. WinRunner also records user interactions and turns them into a script.

LOADRUNNER

A load testing tool from Mercury Interactive Corporation, Mountain View, CA (www.merc-int.com). It simulates thousands of users interacting online in order to test how well a system stands up under a heavy load.

C#

(C Sharp) An object-oriented programming language from Microsoft and ECMA that is based on C++ with elements from Visual Basic and Java. Like Java, C# provides automatic garbage collection, whereas traditional C and C++ do not. C# was created by Microsoft and also standardized by the European Computer Manufacturers Association (ECMA). Microsoft designed C# as its flagship programming language for the .NET environment.

SYSTEMS SOFTWARE C++

An object-oriented version of C that has been widely used to develop enterprise and commercial applications. Created by Bjarne Stroustrup, C++ became popular because it combined traditional C programming with object-oriented programming (OOP) features. Smalltalk and other OOP languages did not provide the familiar structures of conventional languages such as C and Pascal. Microsoft's Visual C++ is the most widely used C++ language.

UNIX

A multiuser, multitasking operating system that is widely used as the master control program in workstations and servers. The Open Group holds the trademark for the UNIX name (spelled in upper case) on behalf of the industry and provides compliance certification to the UNIX standard.

LINUX

Linux is the most popular open source operating system. Its source code is available free of charge; however, for a fee, Linux is distributed with technical support and training from commercial vendors such as Red Hat Software (www.redhat.com) and Novell (www.novell.com). A Linux "distribution" is available as a download or on CD or DVD media, which may comprise from a handful to several hundred applications, tools and utilities. Source code for the Linux kernel as well as the auxiliary programs may also be included.

DSP

Digital Signal Processing) A category of techniques that analyze signals from sources such as sound, weather satellites and earthquake monitors. Signals are converted into digital data and analyzed using various algorithms such as Fast Fourier Transform.

(Digital Signal Processor) A special-purpose CPU used for digital signal processing applications (see definition #2 below). It provides ultra-fast instruction sequences, such as shift and add, and multiply and add, which are commonly used in math-intensive signal processing. DSP chips are widely used in a myriad of devices, including cellphones, sound cards, fax machines, modems, hard disks and digital

TVs. The first DSP chip used in a commercial product was believed to be in the very popular Speak & Spell game, introduced by TI in the late 1970s

CLIENT SERVER COM

Computer Output Microfilm) Creating microfilm or microfiche from the computer. ACOM machine receives print-image output from the computer either online or via tape or disk and creates a film image of each page. The leading vendor of COM machines is Anacomp, Inc., San Diego, CA (www.anacomp.com).

Computer Output Microfilm) Creating microfilm or microfiche from the computer. ACOM machine receives print-image output from the computer either online or via tape or disk and creates a film image of each page. The leading vendor of COM machines is Anacomp, Inc., San Diego, CA (www.anacomp.com).

DCOM

Distributed Component Object Model) Formerly Network OLE, it is Microsoft's technology for distributed objects. DCOM is based on COM, Microsoft's component software architecture, which defines the object interfaces. DCOM defines the remote procedure call that allows those objects to be run remotely over the network. DCOM began shipping with Windows NT 4.0 and is Microsoft's counterpart to CORBA.

DBMS

(Data Base Management System) Software that controls the organization, storage, retrieval, security and integrity of data in a database. It accepts requests from the application and instructs the operating system to transfer the appropriate data. The major DBMS vendors are Oracle, IBM, Microsoft and Sybase. MySQL is a very popular open source product

ORACLE

(Oracle Corporation, Redwood Shores, CA, www.oracle.com) The world's largest database and application software vendor founded in 1977 by Larry Ellison. The Oracle database was the **first** DBMS to incorporate the SQL language and to be ported to a wide variety of platforms. Oracle also offers a variety of development tools.

In the mid-1990s, Oracle was a major promoter of the network computer, forming subsidiary Network Computer, Inc. to define the specifications for the platform. After the turn of the century, the company greatly enhanced its application offerings by acquiring PeopleSoft in 2004 and Siebel Systems in 2005

SQL

Structured Query Language) Pronounced "S-Q-L" or "**see**-quill," a language used to interrogate and process data in a relational database. Originally developed by IBM for its mainframes, all database systems designed for client/server environments support SQL. SQL commands can be used to interactively work with a database or can be embedded within a programming language to interface to a database. Programming extensions to SQL have turned it into a full-blown database programming language, and all major database management systems (DBMSs) support the language.

SYBASE

Sybase Inc., Dublin, CA, www.sybase.com) A software company founded in 1984 that specializes in enterprise infrastructure and integration of platforms, databases and applications. It was originally known for its SQL Server relational DBMS, but expanded its line in 1995 when it acquired Powersoft, makers of the PowerBuilder application development software. Sybase product families include databases, development tools, integration middleware, enterprise portals and mobile and wireless servers.

EMBEDDED SYSTEMS

RTOS

(RealTime Operating System) An operating system designed for use in a realtime computer system

(RTOS) (RealTime Operations) Procedures within an organization that enable information to be distributed to all parties in realtime. It implies that day-to-day activities are integrated with existing information systems so that vital up-to-date information is always available to management, employees and the public as required.

Vx WORKS

A popular realtime operating system for embedded systems from Wind River, Alameda, CA (www.windriver.com). It is used to control a wide variety of products, including network and telecom devices, test and measurement equipment, computer peripherals and consumer products. It is also used in the automotive and aerospace industries for engine control and avionics. Available for a large number of CPU types, applications are created in Wind River's Tornado development environment.

FIRMWARE

A category of memory chips that hold their content without electrical power. Firmware includes flash, ROM, PROM, EPROM and EEPROM technologies. When holding program instructions, firmware can be thought of as "hard software."

BIOS

Basic Input Output System) An essential set of routines stored in a chip that provides an interface between the operating system and the hardware in a PC. The BIOS supports all peripheral technologies including drives as well as internal services such as the realtime clock (time and date). BIOS settings are maintained in a tiny battery-backed memory

QNX

A multiuser, multitasking, realtime operating system for PCs from QNX Software Systems, Ltd., Ottawa, Ontario (www.qnx.com), that is noted for its low-memory requirement and rapid response. Similar to Unix, it has been in use since the early 1980s

MPEG

Moving Pictures Experts Group) An ISO/ITU standard for compressing digital video. Pronounced "**em**-peg," it is the universal standard for digital terrestrial, cable and satellite TV, DVDs and digital video recorders (DVRs).

MPEG uses lossy compression within each frame similar to JPEG, which means pixels from the original images are permanently discarded. It also uses interframe coding, which further compresses the data by encoding only the differences between periodic frames. MPEG performs the actual compression using the discrete cosine transform (DCT) method.

MPEG is an asymmetrical system. It takes longer to compress the video than it does to decompress it in the DVD player, PC, set-top box or digital TV set. As a result, in the early days, compression was performed only in the studio. As chips advanced and became less costly, they enabled digital video recorders, such as Tivos, to convert analog TV to MPEG and record it on disk in realtime.

JPEG

(Joint Photographic Experts Group) An ISO/ITU standard for compressing still images. Pronounced "**jay**-peg," the JPEG format is very popular due to its

variable compression range. JPEGs are saved on a sliding resolution scale based on the quality desired. For example, an image can be saved in high quality for photo printing, in medium quality for the Web and in low quality for attaching to e-mails, the latter providing the smallest file size for fastest transmission over dial-up connections.

SET TOP BOX

The cable TV box that "sits on top" of the TV set. It descrambles the premium channels and provides a tuner for the higher cable numbers that very old TVs did not support. Originally only analog, digital set-top boxes have become widely used for digital services that offer an on-screen program guide. Digital set-top boxes that provide high-definition TV (HDTV) are the latest version.

With satellite TV, a device similar to the set-top box decodes signals for viewing. However, although some call it a "satellite set-top box," it is officially known as a "satellite TV receiver."

DVB

(Digital Video Broadcasting) An international digital television (DTV) standard that is the European and Far Eastern counterpart of the North American ATSC standard.

Administered by the DVB Project within the European Telecommunications Standards Institute (ETSI), DVB uses MPEG-2 for video compression and MPEG-2 and Dolby Digital for audio.

MICROPROCESSORS

A central processing unit (CPU) contained within a single chip. Today, all computer CPUs are microprocessors. The term originated in the 1970s when CPUs up until that time were all comprised of several chips. Thus, when the entire CPU (processor) was miniaturized onto a single chip, the term "micro" processor was coined. Since the turn of the century, the semiconductor manufacturing process has become so sophisticated that not only one, but two or more CPUs, are built on a single chip.

MICROCONTROLLER

A single chip that contains the processor (the CPU), non-volatile memory for the program (ROM or flash), volatile memory for input and output (RAM), a clock and an I/O control unit. Also called a "computer on a chip," billions of microcontroller units (MCUs) are embedded each year in a myriad of products from toys to appliances to automobiles. For example, a single vehicle can use 70 or more microcontrollers.

Microcontrollers come in all sizes and architectures, with the smaller,

commodity chips costing as little as 50 cents in quantities of 10,000.

ADA

A high-level programming language developed by the U.S. Department of Defense along with the European Economic Community and many other organizations. It was designed for embedded applications and process control but is also used for logistics applications. Ada is a Pascal-based language that is very comprehensive.

Ada was named after Augusta Ada Byron (1815-1852), Countess of Lovelace and daughter of the poet Lord Byron and mathematician Annabella Milbanke Byron. Ada also became a mathematician and was the colleague of Charles Babbage, who was developing his Analytical Engine. Some of her programming notes for the machine have survived, giving her the distinction of being the first documented programmer in the world.

MAINFRAMES

AS/400

(Application System/400) The earlier generation and original name of IBM's iSeries and i5 families of midrange business computers. Introduced in 1988, the AS/400 evolved into the iSeries in 2000 and the i5 in 2004. When first introduced, the AS/400 was considered a "minicomputer."

OS/390

The primary operating system used in IBM mainframes. OS/390 was originally the MVS/ESA operating system renamed and repackaged in 1996 with an extensive set of utilities. Although the name MVS is still used to refer to the base control program of OS/390, enhancements in usability and workload balancing have made OS/390 stand apart from its MVS heritage. OS/390 is upward compatible from MVS/ESA 5.2.2, but downward compatibility is not ensured.

z/OS

A mission critical mainframe operating system that extends OS/390 to IBM's zSeries Servers. Although in its first release there are few functional enhancements compared to OS/390 Version 2 Release 10, many more are expected. z/OS, Version 1 Release 1 runs on G5 and G6 Parallel Enterprise Servers, Multiprise 3000 Servers and supports 64-bit real memory addressing on the z900 (64-bit virtual storage is expected). On the G5 and G6, z/OS uses 31-bit addressing and is somewhat restricted. When IBM introduced its zSeries 800 in 2002 for the mid-size market, it introduced a lower-priced

version of z/OS known as "z/OS.e."

OS/400

The operating system for the iSeries family of midrange computers from IBM. Introduced in 1988 for the AS/400 (renamed iSeries in 2000), the OS/400 communicates with the hardware through the Licensed Internal Code (LIC) layer, which includes the device drivers. In 2004, OS/400 was renamed i5/OS to coincide with the eServer i5 models introduced in that same year

VAX

(**V**irtual **A**ddress **eX**tension) A venerable family of 32-bit computers from HP (via Digital and Compaq) introduced in 1977 with the VAX-11/780. VAX models ranged from desktop units to mainframes all running the same VMS operating system, and VAXes could emulate PDP models (Digital's first computers). Large VAX multiprocessing clusters served thousands of users.

COBOL

(**C**ommon **B**usiness **O**riented **L**anguage) A high-level programming language that has been the primary business application language on mainframes and minis. It is a compiled language and was one of the first high-level languages developed. Officially adopted in 1960, COBOL stemmed from FLOWMATIC, a language developed in the mid-1950s by Grace Murray Hopper (later Rear Admiral Hopper) for the UNIVAC I.

COBOL is a very wordy language. Although mathematical expressions can also be written like other programming languages (see example below), its verbose mode is very readable for a novice. For example, **multiply hourly-rate by hours-worked giving gross- pay** is self-explanatory

CICS

(**C**ustomer **I**nformation **C**ontrol **S**ystem) A TP monitor from IBM that was originally developed to provide transaction processing for IBM mainframes. It controls the interaction between applications and users and lets programmers develop screen displays without detailed knowledge of the terminals used. It provides terminal routing, password security, transaction logging for error recovery and activity journals for performance analysis.

CICS has also been made available on non-mainframe platforms including the RS/6000, AS/400 and OS/2-based PCs.

CICS commands are written along with and into the source code of the

applications, typically COBOL, although assembly language, PL/I and RPG are also used. CICS implements SNA layers 4, 5 and 6.

MVS

(**M**ultiple **V**irtual **S**torage) Introduced in 1974, the primary operating system used with IBM mainframes (the others are VM and DOS/VSE). MVS is a batch processing-oriented operating system that manages large amounts of memory and disk space. Online operations are provided with CICS, TSO and other system software.

DB2

(**D**ATABASE 2) A relational DBMS from IBM that was originally developed for its mainframes. It is a full-featured SQL language DBMS that has become IBM's major database product. Known for its industrial strength reliability, IBM has made DB/2 available for all of its own platforms, including OS/2, OS/400, AIX (RS/6000) and OS/390, as well as for Solaris on Sun systems and HP-UX on HP 9000 workstations and servers

IDMSX

(**I**ntegrated **D**ata **M**anagement **S**ystem **E**Xtended) A database management system (DBMS) from Fujitsu Services, formerly ICL, that is widely used on its VME mainframes. It supports journaling, recovery and locking options. A single IDMSX database can contain up to a terabyte of data

IMS

(**I**nformation **M**anagement **S**ystem) An early IBM hierarchical DBMS for IBM mainframes. IMS was widely implemented throughout the 1970s under MVS and continues to be used under z/OS. IMS/DB (IMS/DataBase) is the back end database part, and either IMS/TM (IMS/Transaction Manager) or CICS provides the front end online interaction.

IMS/TM, formerly IMS/DC (IMS/Data Communications), runs each transaction in its own address space and allows for more precise tuning than CICS, which runs all transactions in a region. IMS/TM is also used to access DB2 databases, and Java applications can access IMS databases

TANDEM

(Tandem Computers Inc., Cupertino, CA) A former major manufacturer of fault-tolerant computers founded in 1974 by James Treybig and provider of the early 21st century technology for HP's enterprise computing strategy. Tandem was the first company to address the transaction processing (OLTP) market for online reservations and financial transfers by providing computers

designed from the ground up for fault-tolerant operation. These computers are used in all the major banks, stock exchanges, credit card companies and ATM machines in the world.

Tandem's most significant product was its MIPS-based Himalaya series which ran the NonStop Kernel operating system, compatible with Tandem's Guardian OS. This platform lives on in the NonStop S-series servers from HP, which acquired Tandem's technology via Compaq in 2002. Compaq had purchased Tandem in 1997

Xpeditor

A family of mainframe testing programs from Compuware. It provides the programmer with an assortment of debugging tools for TSO, IMS and other mainframe applications

RPG

Report Program Generator) One of the first program generators designed for business reports, introduced in 1964 by IBM. In 1970, RPG II added enhancements that made it a mainstay programming language for business applications on IBM's System/3x midrange computers. RPG III and RPG IV added more enhancements and have been widely used on the AS/400. RPGLE added the "Integrated Language Environment (ILE)," which enables C, Java and other modules to be integrated into the program.

Until RPGLE, all processing statements were written in strict columnar format. The following RPGLE example changes Fahrenheit to Celsius. The A lines are Data Description Specs (DDS) code. They define a display file and are compiled separately

NETWORKING WINDOWSNT

(Windows **New Technology**) A 32-bit operating system from Microsoft for Intel x86 CPUs. NT is the core technology in Windows 2000 and Windows XP (see [Windows](#)). Available in separate client and server versions, it includes built-in networking and preemptive multitasking. Windows NT was introduced in 1993 as Version 3.1 with the same user interface as Windows 3.1. In 1996, Version 4.0 switched to the Windows 95 desktop and changed some of the dialogs

TCP/IP

(**Transmission Control Protocol/Internet Protocol**) A communications protocol developed under contract from the U.S. Department of Defense to interconnect dissimilar systems. Invented by Vinton Cerf and Bob Kahn, this

de facto Unix standard is the protocol of the Internet and the global standard for communications.

SPX

(**Sequenced Packet EXchange**) The transport layer protocol in the NetWare operating system. Similar to the TCP layer in TCP/IP, it ensures that the entire message arrives intact. SPX uses NetWare's IPX as its delivery mechanism. Application programs use SPX to provide client/server and peer-to-peer interaction between network nodes

SNMP

(**Simple Network Management Protocol**) A widely used network monitoring and control protocol. Data are passed from SNMP agents, which are hardware and/or software

processes reporting activity in each network device (hub, router, bridge, etc.) to the workstation console used to oversee the network. The agents return information contained in a MIB (Management Information Base), which is a data structure that defines what is obtainable from the device and what can be controlled (turned off, on, etc.). Originating in the Unix community, SNMP has become widely used on all major platforms.

CMIP

(**Common Management Information Protocol**) Pronounced "c-mip." A network monitoring and control standard from ISO. CMOT (CMIP over TCP) is a version that runs on TCP/IP networks, and CMOL (CMIP over LLC) runs on IEEE 802 LANs (Ethernet, Token Ring, etc.).

NMS

Also known as [NMS](#), it is an SNMP-based network management software from Novell for monitoring and controlling NetWare networks. NMS was superseded by ManageWise.

LAN

(**Local Area Network**) A communications network that serves users within a confined geographical area. The "clients" are the user's workstations typically running Windows, although Mac and Linux clients are also used. The "servers" hold programs and data that are shared by the clients. Servers come in a wide range

of sizes from Intel-based servers to mainframes. Printers can also be connected to the network and shared

WAN

(Wide Area Network) A long-distance communications network that covers a wide geographic area, such as a state or country. The telephone companies deploy WANs to service large regional areas or the entire nation. Large enterprises have their own private WANs to link remote offices, or they use the Internet for connectivity. The Internet, of course, is the world's largest WAN.

ATM

Asynchronous Transfer Mode) A network technology for both local and wide area networks (LANs and WANs) that supports realtime voice and video as well as data. The topology uses switches that establish a logical circuit from end to end, which guarantees quality of service (QoS). However, unlike telephone switches that dedicate circuits end to end, unused bandwidth in ATM's logical circuits can be appropriated when needed. For example, idle bandwidth in a videoconference circuit can be used to transfer data.

ATM is widely used as a backbone technology in carrier networks and large enterprises, but never became popular as a local network (LAN) topology (see below). ATM is highly scalable and supports transmission speeds of 1.5, 25, 100, 155, 622, 2488 and 9953 Mbps. ATM is also running as slow as 9.6 Kbps between ships at sea. An ATM switch can be added into the middle of a switch fabric to enhance total capacity, and the new switch is automatically updated using ATM's PNNI routing protocol.

MPLS

(MultiProtocol Label Switching) A standard from the IETF for including routing information in the packets of an IP network. MPLS is used to ensure that all packets in a particular flow take the same route over a backbone. Deployed by many telcos and service providers, MPLS can deliver the quality of service (QoS) required to support realtime voice and video as well as service level agreements (SLAs) that guarantee bandwidth. Large enterprises may also use MPLS in their national networks.

Similar to Cisco's tag switching, an MPLS router attaches labels (tags) containing forwarding information to outgoing IP packets. These "label edge routers" (LERs) sit at the edge of the network and perform the complex packet analysis and classification before the packet enters the core of the network. The routers within the core, known as "label switch routers" (LSRs), quickly examine the label and forward the packet per its directions without having to

look up data in tables and compute the forwarding path each time. The edge routers at the receiving end remove the labels.

ISDN

(Integrated Services Digital Network) An international standard for switched, digital dial-up telephone service for voice and data. Analog telephones and fax machines are used over ISDN lines, but their signals are converted into digital by the ISDN terminal adapter (see below).

Although announced in the early 1980s, it took more than a decade before ISDN became widely available. It enjoyed a surge of growth in the early days of the Internet, because it provided the only higher-speed alternative to analog modems in many areas. Still working in many behind-the-scenes applications, ISDN is rarely used for Internet access.

BGP

Border Gateway Protocol) A routing protocol that is used to span autonomous systems on the Internet. It is a robust, sophisticated and scalable protocol that was developed by the Internet Engineering Task Force (IETF). BGP4 supports the CIDR addressing scheme, which has increased the number of available IP addresses on the Internet. BGP was designed to supersede EGP, the original exterior gateway protocol. It is also known as a path vector protocol

RTP

(Rapid Transport Protocol) The protocol used in IBM's High Performance Routing (HPR) system.

(Realtime Transport Protocol) An IP protocol that supports realtime transmission of voice and video. It is widely used for IP telephony and audio and video streaming. An RTP packet rides on top of UDP, the non-reliable counterpart of TCP, and includes timestamping and synchronization information in its header for proper reassembly at the receiving end.

Secure RTP (SRTP) is a version of RTP that provides confidentiality and message authentication.

RTCP

(Realtime Control Protocol): RTCP is a companion protocol to RTP that is used to maintain QoS. RTP nodes analyze network conditions and periodically send each other RTCP packets that report on network congestion.

RTSP

Realtime Streaming Protocol: RTSP is used to control an RTP session at the application layer. It enables functions such as pause, rewind and fast forward to be provided in the user's client software.

RIP

(Raster Image Processor) The hardware and/or software that rasterizes an image for display or printing. RIPs are designed to rasterize a specific type of data, such as PostScript. As desktop computers became more powerful, software RIPs became more appealing than specialized hardware RIPs. Software can be upgraded more easily, and the operation is always speeded up by installing a faster CPU.

(Routing Information Protocol) A simple routing protocol that is part of the TCP/IP protocol suite. It determines a route based on the smallest hop count between source and destination. RIP is a distance vector protocol that routinely broadcasts routing information to its neighboring routers and is known to waste bandwidth. It also has a limit of 15 hops. If a route is advertised as having 16 hops, it is flagged as unreachable. AppleTalk, DECnet, TCP/IP, NetWare and VINES all use incompatible versions of RIP.

(Remote Imaging Protocol) An earlier graphics format from Telegrafix Communications, designed for transmitting graphics over low-speed lines. Using a communications program that supported RIP enabled graphical interfaces to be used on a BBS with respectable performance via modem.

IPv6

(Internet Protocol Version 6) The next generation IP protocol. Started in 1991, the specification was completed in 1997 by the Internet Engineering Task Force (IETF). IPv6 is backward compatible with and is designed to fix the shortcomings of IPv4, such as data security and maximum number of user addresses.

IPv6 increases the address space from 32 to 128 bits, providing for an unlimited (for all intents and purposes) number of networks and systems. It also supports quality of service (QoS) parameters for realtime audio and video. Originally called "IP Next Generation" (IPng), IPv6 is expected to slowly replace IPv4, with the two existing side by side for many years.

IPv6 was officially deployed in July 2004 when ICANN added IPv6 records to its DNS root server for the .jp (Japan) and .kr (Korea) country codes.

SONET

(**Synchronous Optical NETWORK**) A fiber-optic transmission system for high-speed digital traffic. Employed by telephone companies and common carriers, speeds range from 51 Mbps to 40 Gbps.

SONET is an intelligent system that provides advanced network management and a standard optical interface. Specified in the Broadband ISDN (B-ISDN) standard, SONET backbones are widely used to aggregate T1 and T3 lines. The European

counterpart to SONET is the Synchronous Digital Hierarchy, and the term "SONET/SDH" is widely used when referring to SONE

ERP / SAP

Enterprise Resource Planning) An integrated information system that serves all departments within an enterprise. Evolving out of the manufacturing industry, ERP implies the use of packaged software rather than proprietary software written by or for one customer. ERP modules may be able to interface with an organization's own software with varying degrees of effort, and, depending on the software, ERP modules may be alterable via the vendor's proprietary tools as well as proprietary or standard programming languages.

PeopleSoft HRMS

(PeopleSoft, Inc., Pleasanton, CA, www.peoplesoft.com) A software company that specializes in enterprise-wide applications for client/server environments. Initially specializing in human resources, its package offerings today cover the gamut including financial, distribution, manufacturing and supply chain, plus numerous vertical markets. All major databases are supported. Its products are known for their modularity as well as their ease of modification and customization using the PeopleTools development system.

TELECOM

GSM

(**Global System for Mobile Communications**) A digital cellular phone technology based on TDMA that is the predominant system in Europe, but also used worldwide. Developed in the 1980s, GSM was first deployed in seven European countries in 1992. It operates in the 900MHz and 1.8GHz bands in Europe and the 1.9GHz PCS band in the U.S. Based on a circuit-switched system that divides each 200 kHz channel into eight 25 kHz time slots, GSM defines the entire cellular system, not just the TDMA air interface.

GPRS

General Packet Radio Service) An enhancement to the GSM mobile communications system that supports data packets. GPRS enables continuous flows of IP data packet over the system for such applications as Web browsing and file transfer. GPRS differs from GSM's short messaging service (GSM-SMS) which is limited to messages of 160 bytes in length. See [GSM](#).

3G

(3rd Generation) The current generation of data transmission over a cellular network. In CDMA networks such as Verizon Wireless and Sprint, EV-DO is the 3G service. Cingular and other TDMA-based networks support the UMTS technology for 3G, and GPRS is the 3G data service for Vodafone and other GSM carriers.

UMTS

(Universal Mobile Telecommunications System) The European implementation of the 3G wireless phone system. UMTS, which is part of IMT-2000, provides service in the 2GHz band and offers global roaming and personalized features. Designed as an evolutionary system for GSM network operators, multimedia data rates up to 2 Mbps are expected using WCDMA. In the interim, GPRS and EDGE are 2.5G technologies that speed up wireless data for GSM users.

DSP

(Digital Signal Processor) A special-purpose CPU used for digital signal processing applications (see definition #2 below). It provides ultra-fast instruction sequences, such as shift and add, and multiply and add, which are commonly used in math-intensive signal processing. DSP chips are widely used in a myriad of devices, including cellphones, sound cards, fax machines, modems, hard disks and digital TVs. The first DSP chip used in a commercial product was believed to be in the very popular Speak & Spell game.

TDMA

(Time Division Multiple Access) A satellite and cellular phone technology that interleaves multiple digital signals onto a single high-speed channel. For cellular, TDMA triples the capacity of the original analog method (FDMA). It divides each channel into three subchannels providing service to three users instead of one. The GSM cellular system is also based on TDMA, but GSM defines the entire network, not just the air interface.

CDMA

(Code Division Multiple Access) A method for transmitting simultaneous signals over a shared portion of the spectrum. The foremost application of CDMA is the digital cellular phone technology from QUALCOMM that operates in the 800MHz band and 1.9GHz PCS band. CDMA phones are noted for their call quality.

WAP

Wireless Application Protocol) A standard for providing cellular phones, pagers and other handheld devices with secure access to e-mail and text-based Web pages.

Introduced in 1997 by Phone.com (later Openwave Systems), Ericsson, Motorola and Nokia, WAP provides a complete environment for wireless applications that includes a wireless counterpart of TCP/IP and a framework for telephony integration such as call control and phone book access.

SS7

(Signaling System 7) The protocol used in the public switched telephone system (the "intelligent network" or "advanced intelligent network") for setting up calls and providing services. SS7 is a separate signaling network that is used in Class 4 and Class 5 voice switches.

VoIP

(Voice Over IP) A telephone service that uses the Internet as a global telephone network. Many companies, including Vonage, 8x8 and AT&T (CallVantage), typically offer calling within the country for a fixed fee and a low per-minute charge for international. Broadband Internet access (cable or DSL) is required, and regular house phones plug into an analog telephone adapter (ATA) provided by the company or purchased from a third party.

SIGTRAN

(SIGNaling TRANsport) An IP telephony protocol from the IETF that is used to transfer SS7 signals over IP networks. The telco switch sends SS7 signals to a signaling gateway (SG) that converts them into SIGTRAN packets, which travel over IP to the next signaling gateway or to a softswitch if the destination is not another PSTN. SIGTRAN uses the Stream Control Transmission Protocol (SCTP) for reliable transport. See [IP telephony](#) and [SCTP](#).

MGCP/MEGACO

(Media Gateway Control Protocol/MEDIA Gateway Controller) An IP telephony

signaling protocol from the IETF. MGCP was the original protocol, which evolved into MEGACO. Both protocols are designed for implementation in IP phones that are lower cost than SIP or H.323 phones. MGCP/MEGACO requires the use of softswitches for call control and more resembles the telephony model of the circuit-switched PSTN than do SIP and H.323. The softswitch is aware of the entire call throughout its duration (it manages state) and enables operator intervention like the PSTN. MGCP/MEGACO is a combination of the SGCP and IPCD protocols, and many devices that implement MGCP/MEGACO also support SIP and/or H.323.

SIP

(**S**ession **I**nitiation **P**rotocol) An IP telephony signaling protocol developed by the IETF. Primarily used for voice over IP (VoIP) calls, SIP can also be used for video or any media type; for example, SIP has been used to set up multi-player Quake games. With SIMPLE extensions for IM and presence, SIP is also used for instant messaging

H.323

An ITU standard for realtime voice and videoconferencing over packet networks, including LANs, WANs and the Internet. Although H.323 is a very comprehensive standard that supports voice, video, data, application sharing and whiteboarding, the parts relating to audio protocols have been widely used for IP telephony applications.

TMN

(**T**elecommunications **M**anagement **N**etwork) A set of international standards for network management from the ITU. It is used by large carriers such as Sprint, MCI WorldCom and AT&T.

Bluetooth

A wireless personal area network (WPAN) technology from the Bluetooth Special Interest Group (www.bluetooth.com) founded in 1998 by Ericsson, IBM, Intel, Nokia and Toshiba. Bluetooth is an open standard for short-range transmission of digital voice and data that supports point-to-point and multipoint applications. Some of the first Bluetooth applications have been for cellphones, providing a wireless connection to a headset and to an automobile's audio system for hands-free operation

Jini

Pronounced "**gee**-nee." A Java-based distributed computing environment

from Sunin which devices can be plugged into the network and automatically offer their services and make use of other services on the network. Jini creates a "network dialtone" allowing, for example, any PDA or laptop to be plugged in and immediately be able to use printers and other resources. It turns "peripherals into services," so that when a disk drive is plugged in, it becomes a storage service rather than just another disk drive

C, UNIX

Linux

A very popular version of the Unix operating system that runs on a variety of hardware platforms including x86, Itanium, PowerPC and IBM's entire product line. Linux is widely used as a server OS and is gaining ground in the desktop market.

Windows NT

(Windows **New Technology**) A 32-bit operating system from Microsoft for Intel x86 CPUs. NT is the core technology in Windows 2000 and Windows XP (see [Windows](#)). Available in separate client and server versions, it includes built-in networking and preemptive multitasking. Windows NT was introduced in 1993 as Version 3.1 with the same user interface as Windows 3.1. In 1996, Version 4.0 switched to the Windows 95 desktop and changed some of the dialogs (see table below).

VMS

(**Virtual Memory System**) A multiuser, multitasking, virtual memory operating system for the VAX series from Digital. VMS applications run on any VAX from the MicroVAX to the largest unit.

kernel

The fundamental part of a program, typically an operating system, that resides in

memory at all times and provides the basic services. It is the part of the operating system that is closest to the machine and may activate the hardware directly or interface to another software layer that drives the hardware

Tcl/Tk

(**Tool Command Language/ToolKit**) Pronounced "tickle" or "ticklet," it is an interpreted script language that is used to develop a variety of applications, including GUIs, prototypes and CGI scripts. Created for the Unix platform by

John Ousterhout along with students at the University of California at Berkeley, it was later ported to PCs and Macs. Safe-Tcl is an enhanced Tcl interpreter that provides a secure, virus free environment.

Tcl also provides an interface into compiled applications (C, C++, etc.). The application is compiled with Tcl functions, which provide a bi-directional path between Tcl scripts and the executable programs. Tcl provides a way to "glue" program modules together. The Tk part of Tcl/Tk is the GUI toolkit, which is used to create graphical user interfaces. Other languages, including Perl, Python and Scheme, have incorporated Tk as well.

Multithreading

Multitasking within a single program. It allows multiple streams of execution to take place concurrently within the same program, each stream processing a different transaction or message. In order for a multithreaded program to achieve true performance gains, it must be run in a multitasking or multiprocessing environment, which allows multiple operations to take place.

Storage Device

A peripheral unit that holds data such as disk, tape or flash memory card. For a summary of all storage technologies,

iSCSI

(Internet **SCSI**) A protocol that serializes SCSI commands and converts them to TCP/IP

SAN

(**Storage Area Network**) A network of storage disks. In large enterprises, a SAN connects multiple servers to a centralized pool of disk storage.

Compared to managing hundreds of servers, each with their own disks, SANs improve system administration. By treating all the company's storage as a single resource, disk maintenance and routine backups are easier to schedule and control. In

some SANs, the disks themselves can copy data to other disks for backup without any processing overhead at the host

RAID

(**Redundant Array of Independent Disks**) A disk subsystem that is used to increase

performance or provide fault tolerance or both. RAID uses two or more ordinary hard disks and a RAID disk controller. In the past, RAID has also been

implemented via software only.

NAS

(**Network Attached Storage**) A specialized file server that connects to the network. A NAS device contains a slimmed-down operating system and a file system and processes only I/O requests by supporting the popular file sharing protocols, primarily CIFS for Windows and NFS for Unix.

Thank “U” Every One!!

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